

Relations 关系

Rosen 8th ed., ch. 9

- 关系性质
- 关系计算
- 特殊关系

Properties of Relations

关系的性质

- *Reflexive* (自反性)
- *Irreflexive* (反自反性)
- *Symmetric* (对称性)
- *Antisymmetric* (反对称性)
- *Transitive* (传递性)

传递性，例:

传递性 $\Leftrightarrow \forall x \forall y \forall z ((x \in A \wedge y \in A \wedge z \in A \wedge xRy \wedge yRz) \rightarrow xRz)$

判断传递性的典型图例

Y R_1	Y R_2	Y R_3	Y R_4
Y R_5	Y R_6	Y R_7	Y R_8
Y R_9	Y R_{11}	Y R_{12}	

注：独立节点，空关系，恒等关系都是传递的

Combining relations (关系的组合)

- Since relations are by definition sets, they can be combined in any way that sets can be combined, including union, intersection, and set difference.

Example 19

No

Image

Example 19

No

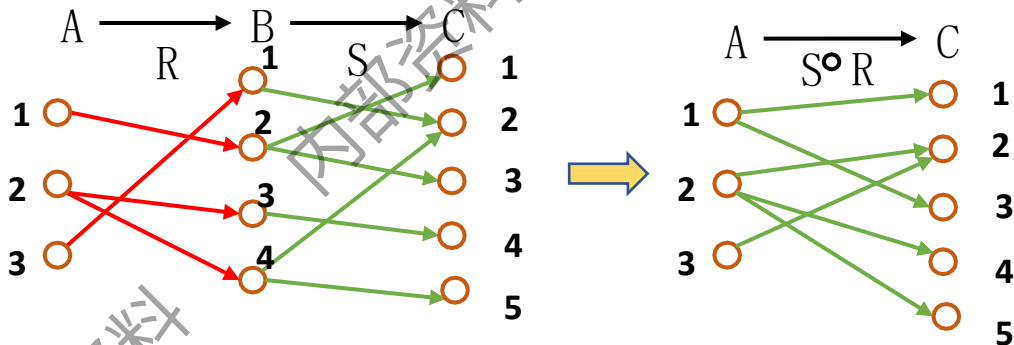
Image

关系合成的方法

例:

No
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1. 有向图法



2. 枚举法

$$R = \{(1,2), (2,3), (2,4), (3,1)\}$$

$$S = \{(1,2), (2,1), (2,3), (3,4), (4,2), (4,5)\}$$

No
Image

§9.4: Closures of Relations

关系闭包

- For any property X , the “ X closure” of a set A is defined as the “smallest” superset of A that has the given property.

关系 R 的 **P 闭包** 就是包含 R 且具有 P 中一切性质的那些关系中的最小的那个。

§9.4: Closures of Relations

- The **reflexive closure** (自反闭包) of a relation R on A is obtained by adding (a,a) to R for each $a \in A$. I.e., it is $R \cup I_A$
- The **symmetric closure** (对称闭包) of R is obtained by adding (b,a) to R for each (a,b) in R . I.e., it is $R \cup R^{-1}$
- The **transitive closure** (传递闭包) or **connectivity relation** of R is obtained by repeatedly adding (a,c) to R for each $(a,b), (b,c)$ in R .
 - I.e., it is

$$R^* = \bigcup_{n \in \mathbb{Z}^+} R^n$$

小结

$r(R) = R \cup I_A$ ----自反闭包

$s(R) = R \cup R^{-1}$ -----对称闭包

$t(R) = R \cup R^2 \cup R^3 \cup \dots \cup R^n$ ——传递闭包

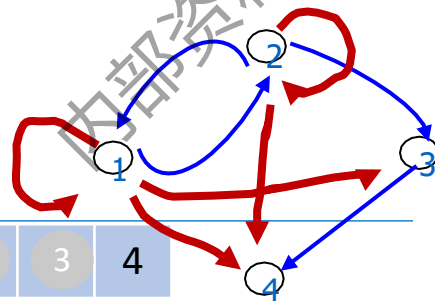
R 是传递的，充要条件 $R^n \subseteq R$

要求：求闭包的几种方法

- 1.枚举法
- 2.有向图方法
- 3.关系矩阵法

Warshall 算法

Example 2



$$W_0 = \begin{bmatrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{bmatrix} = M_R,$$

$$W_1 = \begin{bmatrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{bmatrix}$$

$$W_2 = \begin{bmatrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{bmatrix},$$

$$W_3 = \begin{bmatrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{bmatrix} = W_4 = W^\infty$$

Warshall's Algorithm

另一种方式

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$W_0 = \begin{bmatrix} \underline{0} & \underline{1} & \underline{0} & \underline{0} \\ \underline{1} & \underline{0} & \underline{1} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \underline{1} \\ \underline{0} & \underline{0} & \underline{0} & \underline{0} \end{bmatrix} = M_R, \quad W_1 = \begin{bmatrix} \underline{0} & \underline{1} & \underline{0} & \underline{0} \\ \underline{1} & \underline{1} & \underline{1} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \underline{1} \\ \underline{0} & \underline{0} & \underline{0} & \underline{0} \end{bmatrix}$$

$$W_2 = \begin{bmatrix} \underline{1} & \underline{1} & \underline{1} & \underline{0} \\ \underline{1} & \underline{1} & \underline{1} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \underline{1} \\ \underline{0} & \underline{0} & \underline{0} & \underline{0} \end{bmatrix}, \quad W_3 = \begin{bmatrix} \underline{1} & \underline{1} & \underline{1} & \underline{1} \\ \underline{1} & \underline{1} & \underline{1} & \underline{1} \\ \underline{0} & \underline{0} & \underline{0} & \underline{1} \\ \underline{0} & \underline{0} & \underline{0} & \underline{0} \end{bmatrix} = W_4 = W_\infty$$

几种重要的关系

- 等价关系，等价类，划分
- 偏序关系

p.629, icon at Example 2

#1. Let R be the relation on $\{1, 2, 3, 4\}$ such that

$$R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 3), (4, 4)\}.$$

Find:

- (a) the reflexive closure of R .
- (b) the symmetric closure of R .
- (c) the transitive closure of R .

Hide Solution

Solution:

- (a) $\{(1, 1), (1, 4), (2, 2), (2, 3), (3, 1), (3, 3), (4, 4)\}.$
- (b) $\{(1, 1), (1, 3), (1, 4), (2, 3), (3, 1), (3, 2), (3, 3), (4, 1), (4, 4)\}.$
- (c) $\{(1, 1), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 3), (3, 4), (4, 4)\}.$

p.609, icon at Example 2

#2. Let A be the set of all points in the plane with the origin removed. That is,

$$A = \{(x, y) | x, y \in \mathbf{R}\} - \{(0, 0)\}.$$

Define a relation on A by the rule:

$$(a, b)R(c, d) \leftrightarrow (a, b) \text{ and } (c, d) \text{ lie on the same line through the origin.}$$

- (a) Prove that R is an equivalence relation.
- (b) Describe the equivalence classes arising from the equivalence relation R in part (a).
- (c) If A is replaced by the entire plane, is R an equivalence relation?

See Solution

Solution:

(a) R is reflexive: (a, b) and (a, b) lie on the same line through the origin, namely on the line $y = bx/a$ (if $a \neq 0$), or else on the line $x = 0$ (if $a = 0$).

R is symmetric: if (a, b) and (c, d) lie on the same line through the origin, then (c, d) and (a, b) lie on the same line through the origin.

1

R is transitive: suppose (a, b) and (c, d) lie on the same line L through the origin and (c, d) and (e, f) lie on the same line M through the origin. Then L and M both contain the two distinct points $(0, 0)$ and (c, d) . Therefore L and M are the same line, and this line contains (a, b) and (e, f) . Therefore (a, b) and (e, f) lie on the same line through the origin.

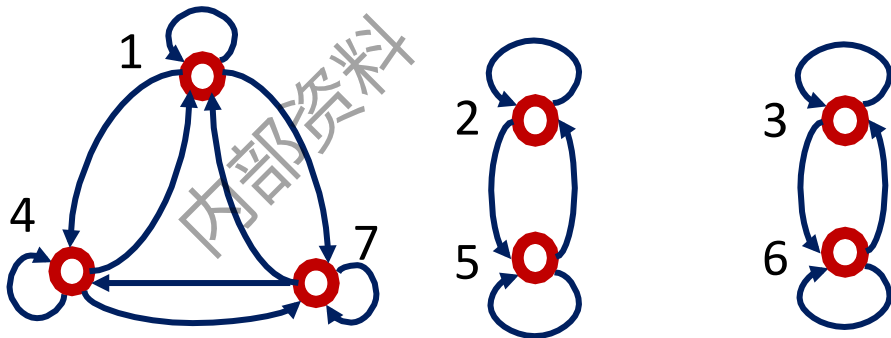
Note: The proof that R is an equivalence relation can be carried out using analytic geometry: if (a, b) and (c, d) lie on the same nonvertical line through the origin, then the slope must equal b/a because the line passes through $(0, 0)$ and (a, b) and the slope must also equal d/c because the line passes through $(0, 0)$ and (c, d) ; thus, $b/a = d/c$, or $ad = bc$. If (a, b) and (c, d) lie on the same vertical line through the origin, then the points must have the form $(0, b)$ and $(0, d)$, and again it must happen that $ad = bc$. Therefore, $(a, b)R(c, d)$ means that $ad = bc$. This equation can be used to verify that R is reflexive, symmetric, and transitive.

- (b) Each equivalence class is the set of points of A on a line of the form $y = mx$ or the vertical line $x = 0$.
- (c) If A is replaced by the entire plane, R is not an equivalence relation. It fails to satisfy the transitive property; for example, $(1, 2)R(0, 0)$ and $(0, 0)R(2, 2)$, but $(1, 2) \not R(2, 2)$ because the line passing through $(1, 2)$ and $(2, 2)$ does not pass through the origin.

等价关系的有向图

画例中模3同余关系R的关系图

$$R = \{ (1,1), (1,4), (1,7), (2,2), (2,5), (3,3), (3,6), (4,1), (4,4), (4,7), (5,2), (6,3), (7,1), (7,4), (7,7) \}$$



Partitions 划分

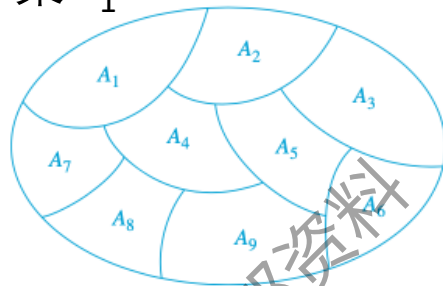
注:

等价关系 R 把 A 的元素分为若干类, 各类之间没有公共元素。 R 是对集合 A 进行的一个划分。

[定义]集合的划分: 把集合 A 分为若干子集 A_1 , A_2 , ..., 满足:

- (1) A_i 非空
- (2) 当 $i \neq j$ 时 $A_i \cap A_j = \Phi$
- (3) $\forall a \in A, \exists i,$
使 $a \in A_i$ ($i = 1, 2, \dots$)

则集合 $P_r(A) = \{A_1, A_2, \dots, A_n, \dots\}$ 称为
 A 的一个划分/partition。



Equivalence classes and quotient set

等价类和商集

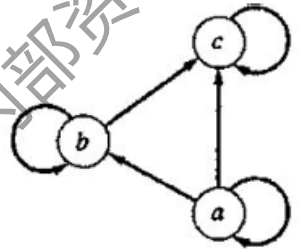
- If R is an equivalence relation on A , then the sets $R(a)$ are traditionally called *equivalence classes* of R , denoted by $[a]$
- The partition $\mathbf{P}$ consists of all equivalence classes of R is denoted by A/R and called
 - the *quotient set* (商集), or
 - the *partition of A induced by R* , or,
 - *A modulo R* .

等价关系 R 确定 A 上的等价类，等价类的集合，称为 A 上关于 R 的商集

§9.6: Partial Orderings --偏序

- Partial Order
 - Linear order (线序)
 - Quasiorder (拟序)
 - Product Partial Order (乘积偏序)
 - Lexicographic Order (词典顺序)
- Hasse Diagram (哈斯图)
- Topological Sorting (拓扑排序)

Hasse Diagrams (哈斯图)



- Hasse Diagram is the digraph of a partial order on a finite set A that
 - deletes all self-cycles (删除自环)
 - eliminates all edges that are implied by the transitive property (取消传递性边)
 - draws the digraph with all edges pointing upward (边方向向上)

Covering Relation

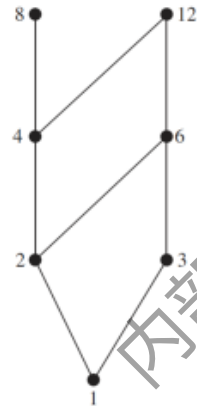
覆盖关系

Let $(S, \preceq)$ be a poset. We say that an element $y \in S$ covers an element $x \in S$ if $x \prec y$ and there is no element $z \in S$ such that $x \prec z \prec y$. The set of pairs (x, y) such that y covers x is called the covering relation of $(S, \preceq)$.

例: $S = \{1, 2, 3, 4, 6, 8, 12\}$, 偏序集
 $(S, |)$ 的覆盖关系?
 $R = \{ (1, 2), (1, 3), \dots \}$

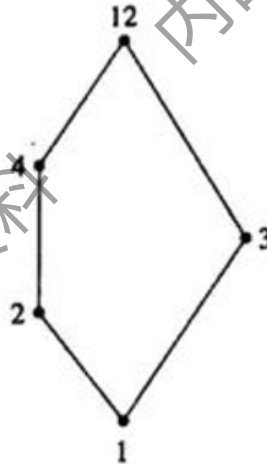
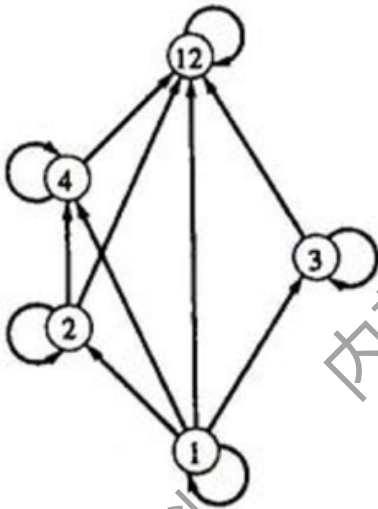
$(2, 4)$ $(2, 8)$ $(2, 6)$
 ? ? ?

求关系图的传递闭包?

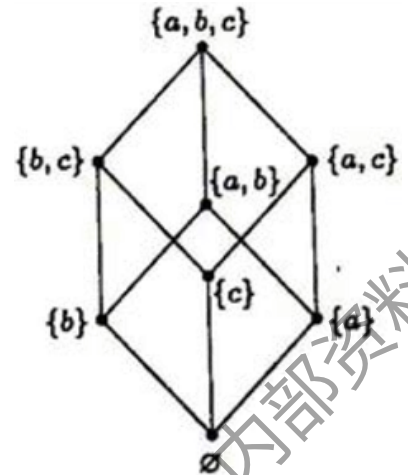


Hasse Diagram

• $(\{1, 2, 3, 4, 12\}, |)$



• $(P(\{a, b, c\}), \subseteq)$



偏序集的重要元素

Extremal elements
(极值元素)

Extremal elements

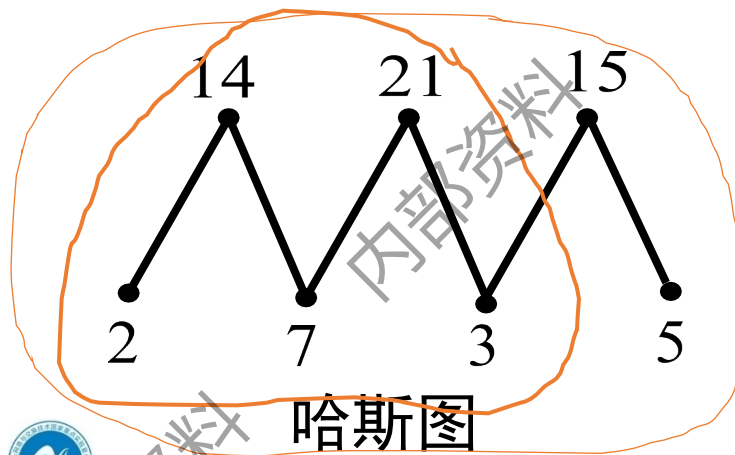
- *maximal element* (极大元)
- *minimal element* (极小元)
- *greatest element* (最大元)
 - denoted by 1 and often called the *unit element* (单位元)
- *least element* (最小元)
 - denoted by 0 and often called the *zero element* (零元)

例: $P = \{ 2, 3, 5, 7, 14, 15, 21 \}$, 整除关系, 子集

$A = \{ 2, 7, 3, 21, 14 \}$

极大元: $\{ 14, 21 \}$

极小元: $\{ 2, 7, 3 \}$



哈斯图

当 $A = P$ 时

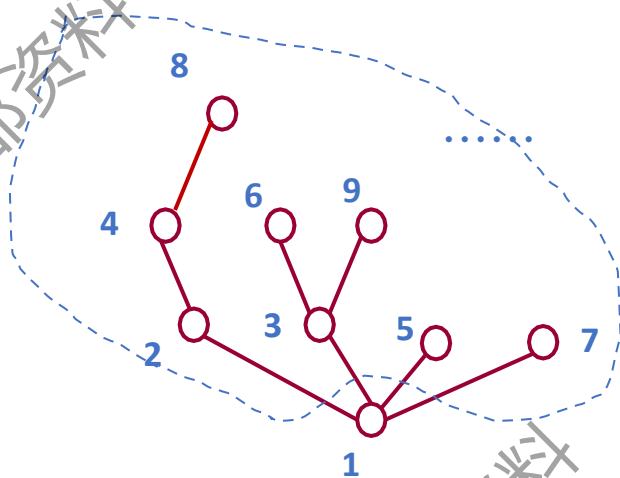
极大元为 14, 15, 21

极小元为 2, 3, 5, 7.

例：自然数集合 N 上的整除关系“ $|$ ”
构成偏序集 $(N, |)$
 $A = \{2, 3, 4, 5, 6, 7, 8, 9\}$

A 中极大元：8, 6, 9, 5, 7

A 中极小元：2, 3, 5, 7



注意：极大元、极小元并不唯一，且同一元素
即可能是极大元，又可能是极小元

最大元 (最小元)

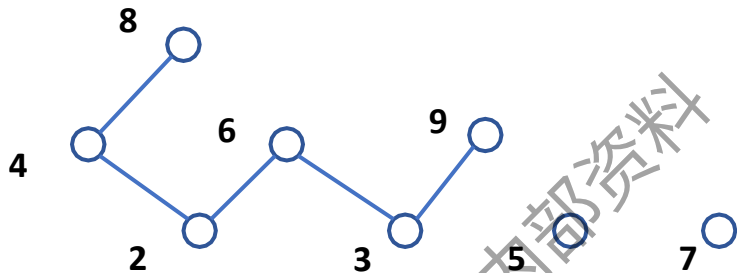
定义：偏序集 $(P, \leq)$ 中，子集 $A \subseteq P$ ，若 $a \in A$ ，对 $\forall b \in A$ ，都有 $b \leq a$ 则 a 为 A 的最大元；若 $a \in A$ ，对 $\forall b \in A$ ，都有 $a \leq b$ 则 a 为 A 的最小元。

注意：最大元和最小元在子集 A 中找。子集 A 的最大元 (最小元) 如果存在，则与 A 中的任意元素必须可比

例： $(N, |)$ 的哈斯图

$A = \{2, 3, 4, 5, 6, 7, 8, 9\}$

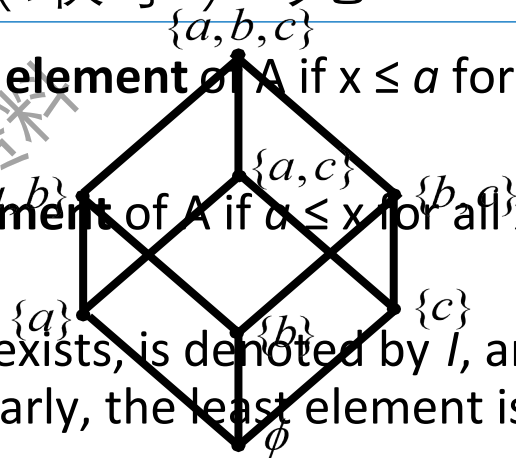
无最大元，无最小元



Greatest (least) element

集合A上的最大（最小）元

- An element $a \in A$ is called a **greatest element** of A if $x \leq a$ for all $x \in A$.
- An element $a \in A$ is called a **least element** of A if $a \leq x$ for all $x \in A$.
- The greatest element of a poset, if it exists, is denoted by 1 , and is often called the **unit** element. Similarly, the least element is called **zero** element.



➤ 如果 $\exists a \in A$, $\forall x \in A$, 都有 $x \leq a$, 则 a 为最大元, 用 **1** 表示---称为单位元

➤ 如果 $\exists a \in A$, $\forall x \in A$, 都有 $a \leq x$, 则 a 为最小元, 用 **0** 表示---称为零元

Lub (最小上界)

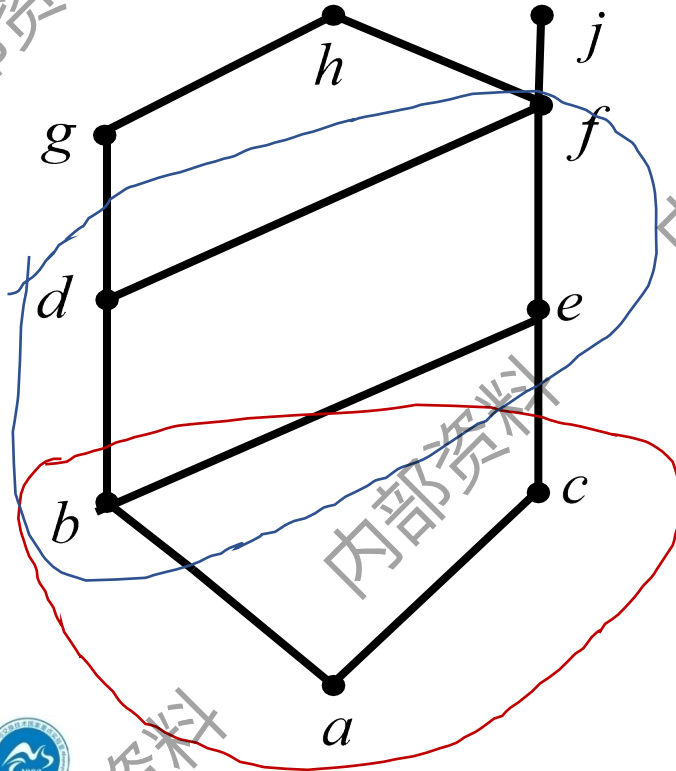
Glb (最大下界)

- poset A and a subset $B \subseteq A$. (B 是 A 的子集)

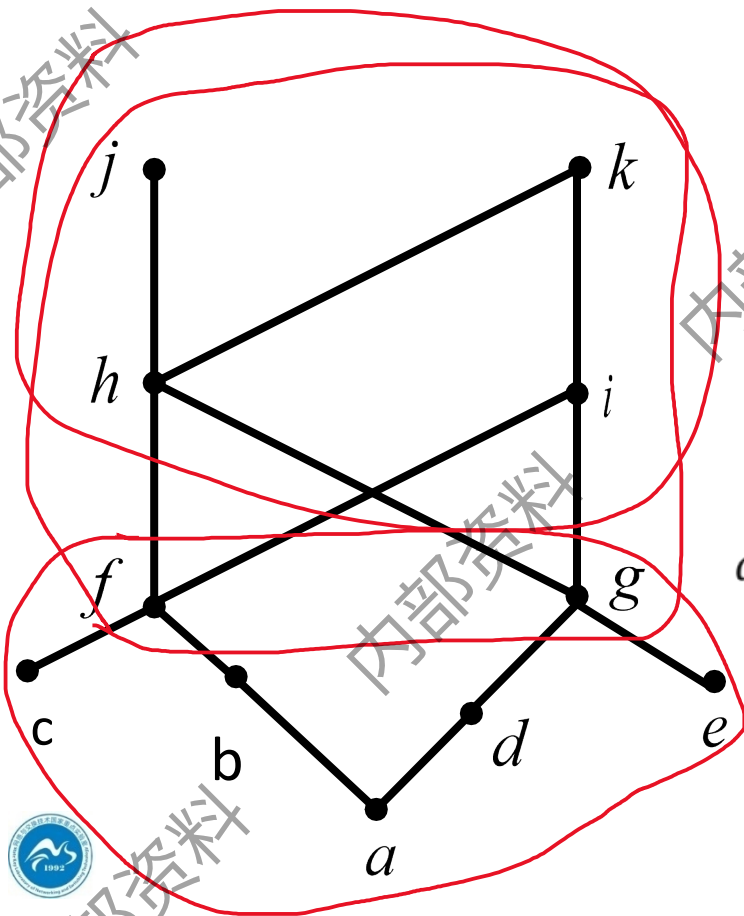
注意:

- An element $a \in A$ is called a **upper bound** (上界) of B if $b \leq a$ for all $b \in B$.
上界和上界都在全集 A 中找, 不限于子集 B
- An element $a \in A$ is called a **lower bound** (下界) of B if $a \leq b$ for all $b \in B$.
- An element $a \in A$ is called a **least upper bound** (LUB) of B if a is an upper bound of B and $a \leq a'$, whenever a' is an upper bound of B .
上界和下界可以不唯一, 也可以不存在
- 如果存在最小上界和最大下界, 则一定唯一
- An element $a \in A$ is called a **greatest lower bound** (GLB) of B if a is an lower bound of B and $a' \leq a$, whenever a' is an lower bound of B .

Example 18 : Find the lower and upper bounds of
 $\{a, b, c\}, \{j, h\}, \{b, e, d, f\}$



LUB and GLB of $\{b, d, g\}$?



$$B = \{a, b, c, d, e, f, g\}$$

h, i, j, k 为 B 的上界

f, g 为 $B' = \{j, h, k, i\}$ 的下界

a, b, c, d, e 也为 $B' = \{j, h, k, i\}$ 的下界

a 是 $\{f, g, h, i, j, k\}$ 的下界

b, c, d, e 不是 $\{f, g, h, i, j, k\}$ 的下界

Theorem 1,2,3

- 总结
- Let $(A, \leq)$ be a poset
 - If A is a finite nonempty, then A has at least **one** maximal element and at least **one** minimal element. (有限非空集合极大元/极小元各至少一个)
 - A has at most **one** greatest element and at most **one** least element. (最大元/最小元各最多一个)
 - A subset B of A has at most **one** LUB and at most **one** GLB. (子集 B 的最小上界/最大下界各最多一个)

思考：什么情况可比，什么情况不可比？

p.650, icon at Example 1

#1. Let $A = \{(x, y) | x, y \text{ integers}\}$. Define a relation R on A by the rule

$$(a, b)R(c, d) \leftrightarrow a \leq c \text{ or } b \leq d.$$

Determine whether R is a partial order relation on A .

Hide Solution

Solution:

R is reflexive: $(a, b)R(a, b)$ for all elements (a, b) because $a \leq a$ or $b \leq b$ is always true.

R is not antisymmetric: For example, $(1, 4)R(3, 2)$ because $1 \leq 3$, and $(3, 2)R(1, 4)$ because $2 \leq 4$. But $(1, 4) \neq (3, 2)$.

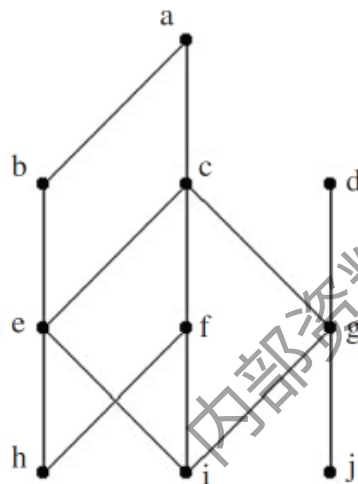
R is not transitive: For example, $(1, 4)R(3, 2)$ because $1 \leq 3$, and $(3, 2)R(0, 3)$ because $2 \leq 3$. But $(1, 4) \not R(0, 3)$ because $1 \not\leq 0$ and $4 \not\leq 3$.

Therefore, R is not a partial order relation because R is neither antisymmetric nor transitive.

p.658, icon at Example 20

#1. Referring to this Hasse diagram of a partially ordered set, find the following:

- (a) all upper bounds of $\{d, e\}$.
- (b) the least upper bound of $\{d, e\}$.
- (c) all lower bounds of $\{a, e, g\}$.
- (d) the greatest lower bound of $\{a, e, g\}$.
- (e) greatest lower bound of $\{b, c, f\}$.
- (f) least upper bound of $\{h, i, j\}$.
- (g) greatest lower bound of $\{g, h\}$.
- (h) least upper bound of $\{f, i\}$.



Solution:

- (a) There are no upper bounds of $\{d, e\}$.
- (b) Because there are no upper bounds of $\{d, e\}$, there is no least upper bound of $\{d, e\}$.
- (c) The only lower bound of $\{a, e, g\}$ is i .
- (d) The glb of $\{a, e, g\}$ is the only lower bound of $\{a, e, g\}$, namely i .
- (e) Both h and i are lower bounds of $\{b, c, f\}$. But there is no greatest lower bound.
- (f) Both a and c are upper bounds of $\{h, i, j\}$. The element c is the least upper bound.
- (g) There is no lower bound of $\{g, h\}$. Hence there is no greatest lower bound.
- (h) The elements a, c , and f are upper bounds of $\{f, i\}$. The element f is the least upper bound.

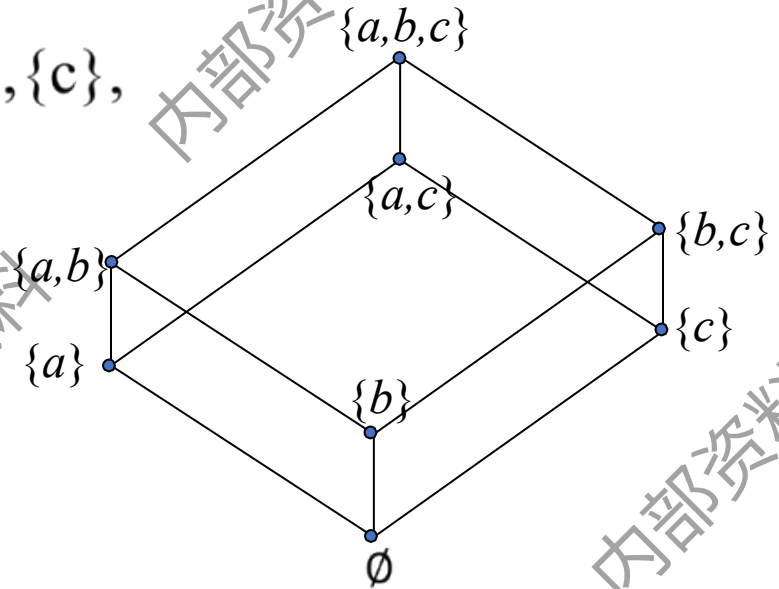
Lattices (格)

Lattice (格)

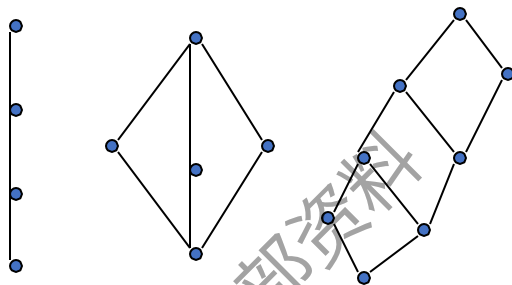
- A *lattice* is a poset in which every subset $\{a,b\}$ consisting of two elements has a least upper bound (LUB) and a greatest lower bound (GLB).
 - denote $\text{LUB}(\{a,b\})$ by $a \vee b$ and call it the *join* (并) of a and b .
 - denote $\text{GLB}(\{a,b\})$ by $a \wedge b$ and call it the *meet* (交) of a and b .
- 若偏序集中任意两个元素 a, b 都有最小上界 ($\text{LUB}\{a,b\}$) 和最大下界 ($\text{GLB}\{a,b\}$), 则偏序集为格
- $\text{LUB}(\{a,b\}) = a \vee b, \quad \text{GLB}(\{a,b\}) = a \wedge b$

Example

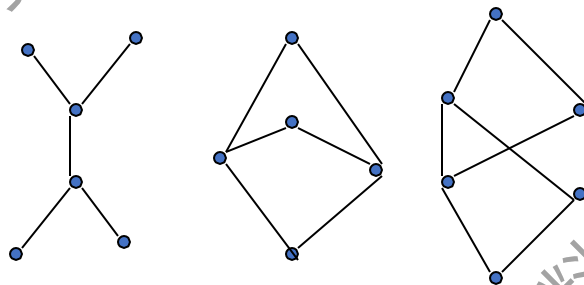
- $S = \{a, b, c\}$
- $P(S) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$



例:判断是否为格



格



不是格

SORT Algorithm

拓扑排序算法

- For finding a topological sorting of a finite poset $(A, \leq)$.
- Step1 Choose a minimal element a of A .
- Step2 Make a the next entry of SORT and replace A with $A - \{a\}$.
- Step3 Repeat steps 1 and 2 until $A = \{\}$.
- End of Algorithm

每一步从 A 中删除1个极小元，直至 $A = \{\}$

- Example 26
- Find a compatible total ordering for the poset
- $(\{1, 2, 4, 5, 12, 20\}, |)$

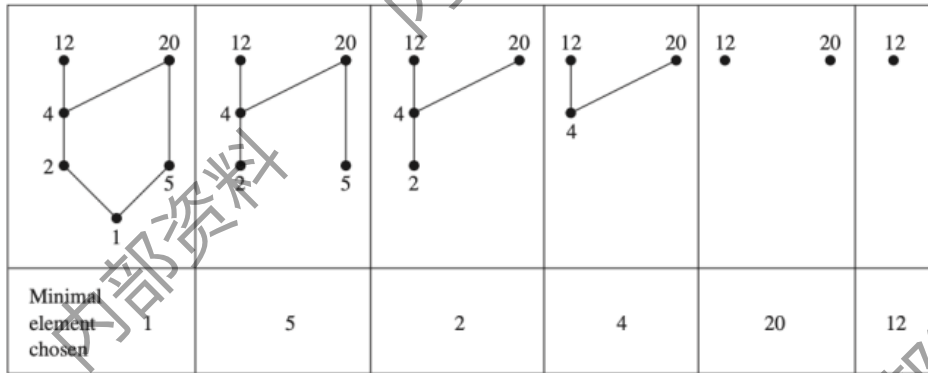


FIGURE 9 A Topological Sort of $(\{1, 2, 4, 5, 12, 20\}, |)$.

全序关系: $1 < 5 < 2 < 4 < 20 < 12$.

Semigroup and Groups

9.1 二元运算/Binary Operations Revisited

9.2 半群/Semigroups

9.3 乘积半群与商半群/Products and Quotients of Semigroups

9.4 群Groups

9.5 乘积群与商群Products and Quotients of Groups

Subsemigroup – 子半群

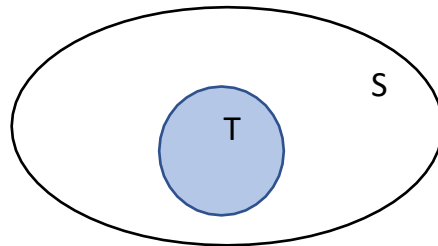
Let

$(S, *)$ be semigroup and

$T \subseteq S$

if T is **closed** under the operation $*$, then $(T, *)$ is called **subsemigroup** of $(S, *)$.

- 子集
- 运算封闭
- **结合性?**



Submonoid – 子独异点

Let

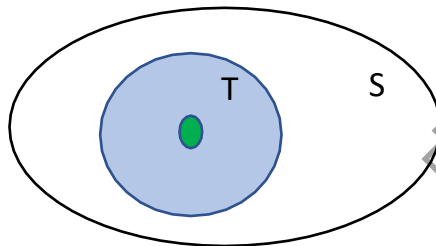
$(S, *)$ be a monoid with identity e , and

$T \subseteq S$.

if T is **closed** under the operation $*$ and $e \in T$, then

$(T, *)$ is called a **submonoid** of $(S, *)$.

- 子集
- 运算封闭
- 幺元属于 T



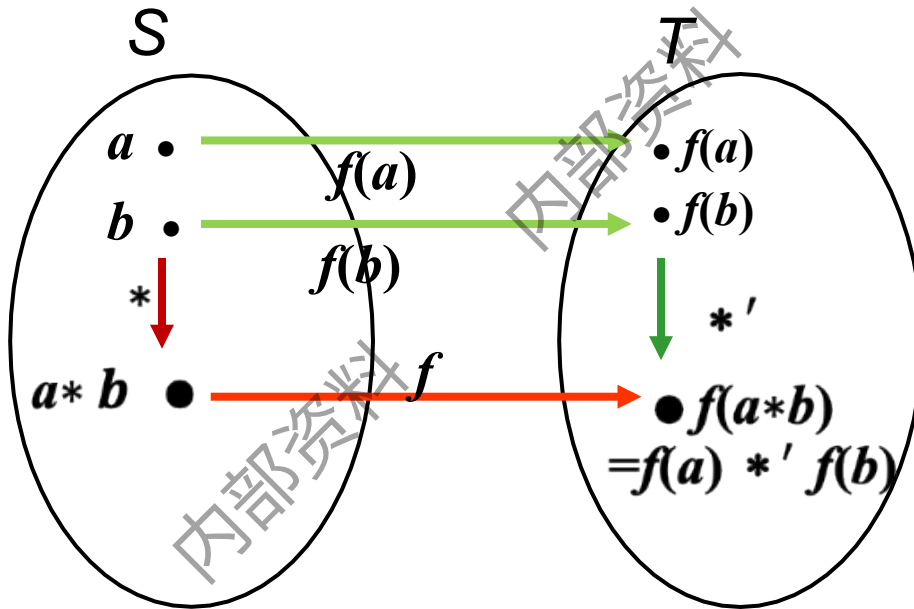
Isomorphism – 同构

- Let $(S, *)$ and $(T, *')$ be two semigroups.
- A function $f: S \rightarrow T$
- is called an *isomorphism* from $(S, *)$ to $(T, *')$
- if it is a one-to-one correspondence from S to T , and $f(a * b) = f(a) *' f(b)$ for all a and b in S .

设 $(S, *)$ 和 $(T, *')$ 为两个半群，如果函数 $f: S \rightarrow T$ 是 S 到 T 的一一对应，并且对 S 中所有的 a 和 b 有

$$f(a * b) = f(a) *' f(b)$$

则称函数 $f: S \rightarrow T$ 是从 $(S, *)$ 到 $(T, *')$ 的一个同构



Homomorphism – 同态

➤ Let $(S, *)$ and $(T, *')$ be two semigroups.

An everywhere-defined function $f: S \rightarrow T$ is called a **homomorphism** from $(S, *)$ to $(T, *')$

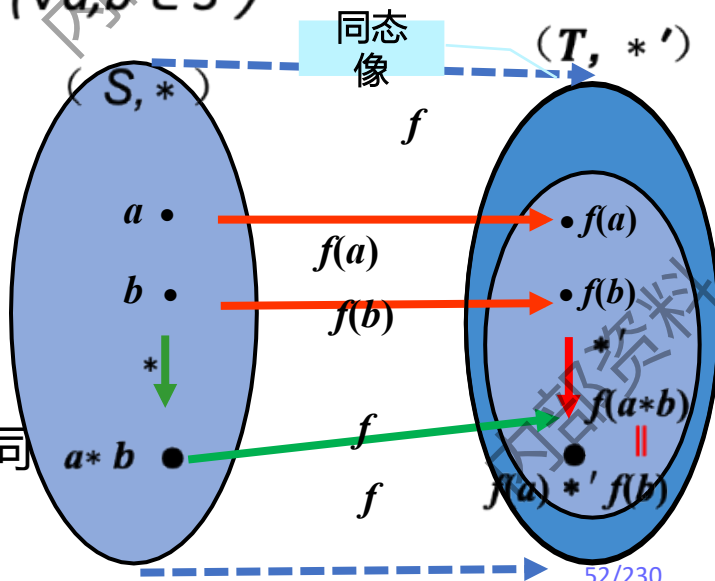
If $f(a * b) = f(a) *' f(b) \ (\forall a, b \in S)$

for all a and b in S .

➤ If f is also onto, we say that T is a **homomorphic image** (同态像) of S .

如果 f 是满射的, 则 T 是 S 的同态像

单同态, 自同态



9.3 Products and Quotients of Semigroups

乘积半群、商半群

Products and Quotients of semigroups

- Products semigroups (积半群)
- Congruence relation (同余关系)
- Quotients of semigroups (商半群)
- Fundamental Homomorphism Theorem
- (基本同态原理)



积半群定义

- If $(S, *)$ and $(T, *')$ are semigroups, then $(S \times T, *'')$ is a semigroup, where $*''$ is defined by
 - $(s_1, t_1) *'' (s_2, t_2) = (s_1 * s_2, t_1 *' t_2)$
 - Proof: omitted.
- What about **monoid**?
- if S and T are monoids with identities e_S and e_T , respectively then $S \times T$ is a monoid with **identity** (e_S, e_T) .



Congruence relation – 同余关系

An equivalence relation R on the semigroup $(S, *)$ is called a **congruence (同余) relation** if $a R a'$ and $b R b'$ imply $(a * b) R (a' * b')$

等价关系不一定是同余关系，因为同余必须与定义在 S 上二元运算密切相关

Theorem 2---商半群

- Let R be a **congruence relation** on the semigroup $(S, *)$.
- Consider the relation $\odot$ from $S/R \times S/R$ to S/R in which the ordered pair $([a], [b])$ is, for a and b in S , related to $[a * b]$. (构造商半群 $(S/R, \odot)$)
 - $\odot([a], [b]) = [a] \odot [b] = [a * b]$, is a function from $S/R \times S/R$ to S/R
 - $(S/R, \odot)$ is a semigroup. S/R is called the **quotient semigroup** (商半群)

R 是半群 $(S, *)$ 上的同余关系

$\odot$ 是 $S/R \times S/R$ to S/R 的函数



Fundamental Homomorphism Theorem 4– 同态基本定理

- Let

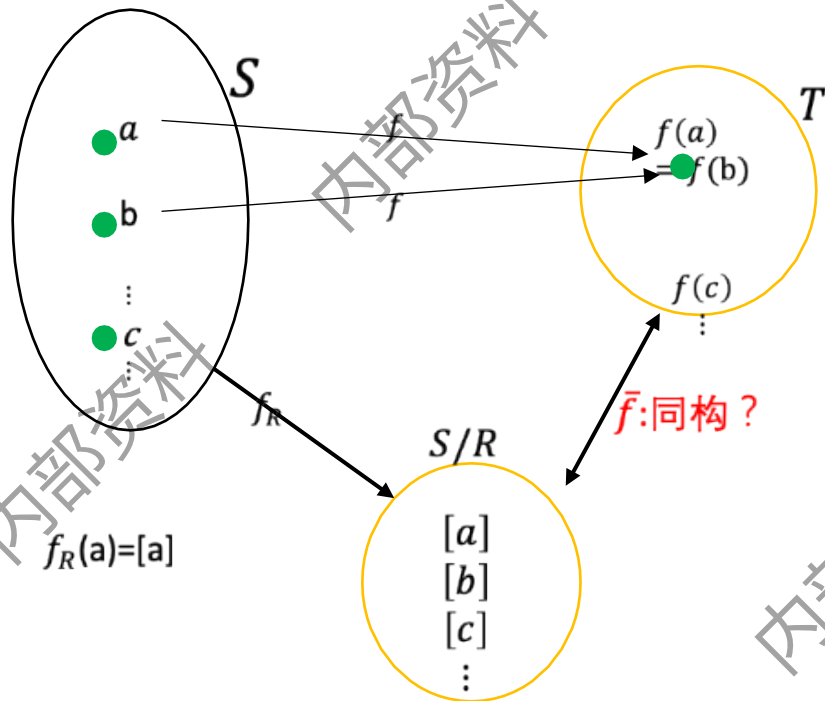
- $f: S \rightarrow T$ be a **homomorphism** of the semigroup $(S, *)$ **onto** the semigroup $(T, *)$.
- R be the relation on S defined by
 - $a R b$ **if and only if** $f(a) = f(b)$, for a and b in S .

- Then

- (1) R is a congruence relation.
- (2) $(T, *)$ and the quotient semigroup $(S/R, \odot)$ are isomorphic.



aRb



9.4 Groups – 群

Groups

- Definition (定义)
- Abelian (阿贝尔群)
- Properties of groups (群性质)
- Finite groups (有限群)
- Subgroups (子群)

群定义

A group $(G, *)$ is a monoid, with identity e , that has the additional property that for every element $a \in G$ there exists an element $a' \in G$, such that

$$a * a' = a' * a = e.$$

group

A **group** $(G, *)$ is a set together with a binary operation $*$ on G such that, for any elements a, b , and c in G . $\forall a, b, c \in G$

➤ $a * b \in G$ -- 封闭

➤ $(a * b) * c = a * (b * c)$ -- 结合

a unique element e in G such that

➤ $a * e = e * a$ -- 单位元

an element $a' \in G$, called an inverse of a and written as a^{-1} , such that

➤ $a * a' = a' * a = e$ or

$aa' = a'a = e$ or $aa^{-1} = a^{-1}a = e$ ----- 每个元素有逆元

群的常用术语

平凡群:只含单位元的群 $\{e\}$

交换群:Abel群

有限群与无限群

群 G 的阶, G 的基数,通常有限群记为 $|G|$

元素 a 的 n 次幂

$$a^n = \begin{cases} e & n = 0 \\ a^{n-1}a & n > 0 \\ (a^{-1})^n & n < 0 \end{cases}$$

元素 a 的阶 $|a|$,使得 $a^k = e$ 成立的最小正整数 k

注:有限群的元素都是有限阶,是群的阶的因子,

反之,元素都是有限阶的群不一定是有限群

Example 4

- Let
 - G be the set of all nonzero real numbers and
 - $a * b = ab/2$.
- Show
 - $(G, *)$ is an Abelian group.

Example 4 - Solution

- $*$ is a **binary operation** – 运算封闭
 - If a and b are elements of G ,
 - then $ab/2$ is a nonzero real number and
 - hence is in G .
- **associativity** -- 可结合
 - $(a * b) * c = (ab/2) * c = (ab)c/4$
 - $a * (b * c) = a * (bc/2) = a(bc)/4 = (ab)c/4$.
 - $*$ is associative.

Example 4 - Solution

- 2 is the identity (有单位元)
 - $a * 2 = (a)(2)/2 = a = (2)(a)/2 = 2 * a.$
- $a' = 4/a$ is an inverse of a ---逆元
 - $a * a' = a * 4/a = a(4/a)/2 = 2 = (4/a)(a)/2$
 - $= (4/a) * a = a' * a.$
- Abelian
 - $a * b = ab/2 = ba/2 = b * a$
- So, G is an Abelian group.

群的性质小结

幂运算规则:

1. $(a^{-1})^{-1} = a$

2. $(ab)^{-1} = b^{-1}a^{-1}$ 可以推广到 n 个元素的积的逆。

3. $(a^n)^{-1} = (a^{-1})^n$

4. $a^m a^n = a^{m+n}$

5. $(a^m)^n = a^{mn}$

6. 若 G 为Abel群, 则 $(ab)^n = a^n b^n$

群的性质

6. 群中除了单位元 e 外，没有其他的幂等元。

(假设 $a*a=a$ ，因为 $a*e=a$ ，根据消去律 $a=e$)

消去律:

$$1. \quad ab=ac \quad \Rightarrow \quad b=c \quad \text{--左消去}$$

$$2. \quad ba=ca \quad \Rightarrow \quad b=c \quad \text{--右消去}$$

Example 5

- Let $B = \{0, 1\}$, and let $+$ be the operation defined on B as follows:

$+$	0	1
0	0	1
1	1	0

- Then B is a group.

2阶群每个元素是它自身的逆



Group of symmetries of the triangle 三角形对称群

- Let $S_3 = \{f_1, f_2, f_3, g_1, g_2, g_3\}$ and introduce the operation $*$, *followed by*, on the set S_3

$*$	f_1	f_2	f_3	g_1	g_2	g_3
f_1	f_1	f_2	f_3	g_1	g_2	g_3
f_2	f_2	f_3	f_1	g_3	g_1	g_2
f_3	f_3	f_1	f_2	g_2	g_3	g_1
g_1	g_1	g_2	g_3	f_1	f_2	f_3
g_2	g_2	g_3	g_1	f_3	f_1	f_2
g_3	g_3	g_1	g_2	f_2	f_3	f_1

$*$ 表示合成运算, 如 $f_2 * g_2$ 表示, 先进行 g_2 变换, 再进行 f_2 变换

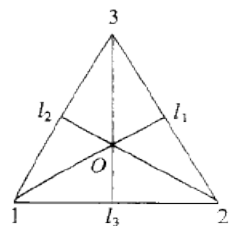


Figure 9.3

$$f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, f_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, f_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$g_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, g_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, g_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$



Subgroup – 子群

内部资料

- Let H be a subset of a group G such that
 - (a) The identity e of G belongs to H .
 - (b) If a and b belong to H , then $ab \in H$.
 - (c) If $a \in H$, then $a^{-1} \in H$.
- Then H is called a **subgroup** of G .

判断子群的方法一: $H \subseteq G$

➤ $e \in H$

➤ If $\forall a, b \in H$, then $ab \in H$ —— 封闭

➤ If $a \in H$, then $a^{-1} \in H$ —— 都有逆

Subgroup – 子群

判断子群的方法二: 非空 $H \subseteq G$

$$\triangleright \forall a, b \in H, a b^{-1} \in H$$

证: 1. H 非空, $\exists b \in H$

• 2. $\exists b \in H \Rightarrow b b^{-1} \in H \Rightarrow e \in H$ (根据 $a b^{-1} \in H$)

3. $\forall a \in H \Rightarrow e a^{-1} \in H \Rightarrow a^{-1} \in H$

4. $\forall a, b \in H \Rightarrow a b^{-1} \in H \Rightarrow a (b^{-1})^{-1} \in H$
 $\Rightarrow a b \in H$

含单位元

证都有逆元

证封闭



5-57 设 f 和 g 都是群 $\langle G_1, \star \rangle$ 到群 $\langle G_2, * \rangle$ 的同态, 证明:
 $\langle C, \star \rangle$ 是 $\langle G_1, \star \rangle$ 的一个子群, 其中

$$C = \{x | x \in G_1 \text{ 且 } f(x) = g(x)\}$$

【5-8.(11)】

证明 由定义可知 $C \subseteq G_1$ 。

对于任意的 $a, b \in C$, 有 $f(a) = g(a), f(b) = g(b)$ 。

因为 f 和 g 都是同态映射, 所以必有

$$f(b^{-1}) = f(b)^{-1}, g(b^{-1}) = g(b)^{-1}$$

现因 $f(b) = g(b)$, 故有 $f(b)^{-1} = g(b)^{-1}$, 即有 $f(b^{-1}) = g(b^{-1})$ 。由

此可得 $f(a \star b^{-1}) = f(a) * f(b^{-1}) = g(a) * g(b^{-1}) = g(a \star b^{-1})$, 所以 $a \star b^{-1} \in C$ 。

因此, $\langle C, \star \rangle$ 是 $\langle G_1, \star \rangle$ 的子群。



Examples9

- Let G be a group. Then G and $H = \{e\}$ are subgroups of G , called the *trivial subgroups* (平凡子群) of G .
- G 和 $H = \{e\}$ 是 G 的平凡子群

Examples10

- Consider S_3 , the group of symmetries of the equilateral triangle. It is easy to verify that $H = \{f_1, f_2, f_3\}$ is a subgroup of S_3 .

*	f_1	f_2	f_3	g_1	g_2	g_3
f_1	f_1	f_2	f_3	g_1	g_2	g_3
f_2	f_2	f_3	f_1	g_3	g_1	g_2
f_3	f_3	f_1	f_2	g_2	g_3	g_1
g_1	g_1	g_2	g_3	f_1	f_2	f_3
g_2	g_2	g_3	g_1	f_3	f_1	f_2
g_3	g_3	g_1	g_2	f_2	f_3	f_1

生成子群的一种方法： 元素 a 的幂生成的子群

- Let
 - G be a group and $a \in G$.
- Define
 - a^n as $aa \dots a$ (n factors), for $n \in \mathbb{N}$ and
 - a^0 as e .
 - a^{-n} as $a^{-1}a^{-1} \dots a^{-1}$ (n factors).
- if n and m are any integers, then $a^n a^m = a^{n+m}$.
- It is easy to show that
 - $H = \{ a^i \mid i \in \mathbb{Z} \}$ is a subgroup of G . (generated by a)

1.非空

2.封闭

3.有单位元, $a^0 = e = aa^{-1} \in H$

4.存在逆元, $a^i = a^i a^{-i} = e, a^{-i} \in H$

5.可结合

元素 a 的幂生成的子群 (续)

- 例： $(\mathbb{Z}, +)$, 单位元0
- 比如生成元 $2 \in \mathbb{Z}$, 由2 可生成偶数群
- $(\mathbb{T}, +)$ 是 $\mathbb{Z}$ 的子群。
- $2^0 = 0, 2^1 = 2, 2^2 = 4, 2^3 = 6, \dots$
- $2^{-1} = -2, \dots$
- 由3 可生成子群?
-
- 由 1可生成子群?

Cyclic group – 循环群

- Recall the subgroup generated by a . Now Check Table 9.7



Table 9.5

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Table 9.6

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

Table 9.7

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

Table 9.8

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

生成元 a 生成 G 的所有元素

- 循环群至少含一个生成元

循环群一续

实例:

$(\mathbb{Z}, +)$ 为无限循环群？

生成元1和-1

$(\mathbb{Z}_n, \oplus)$, 为n阶循环群

$(\mathbb{Z}_6, \oplus)$, [1]和 [5] 是生成元;

生成子群阶分别为1,2,3,6

由[0]生成1阶子群 $\{[0]\}$

由[1]生成6阶子群 $\mathbb{Z}_6$

由[2]生成3阶子群 $\{[0], [2], [4]\}$

由[3]生成2阶子群 $\{[0], [3]\}$

Theorem 5—群同态的特性

- Let

- $(G, *)$ and $(G', *')$ be two groups, and
- $f: G \rightarrow G'$ be a homomorphism from G to G' .

- Then

- (a) If e is the identity in G and e' is the identity in G' , then $f(e) = e'$.
- (b) If $a \in G$, then $f(a^{-1}) = (f(a))^{-1}$.
- (c) If H is a subgroup of G , then $f(H) = \{f(h) \mid h \in H\}$ is a subgroup of G' .

$$H \subseteq G \Rightarrow f(H) \subseteq G':$$

$$f: G \rightarrow G' \Rightarrow f(a * a^{-1}) = f(a) *' f(a^{-1})$$

$$\Rightarrow f(e) = f(a) *' f(a^{-1}) \Rightarrow f(a) = f(a^{-1})^{-1}$$

Example 17

- The groups with multiplication Tables 9.6, 9.7, and 9.8 are isomorphic

Table 9.5

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Table 9.6

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

Table 9.7

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

Table 9.8

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

- The group with multiplication Table 9.5 is called the *Klein 4 group* (克莱因四元素群) and it is denoted by V .

- Tables 9.5 and 9.6 不同构?

- Observe that if x is any element in the group determined by Table 9.5, then $x^2 = e$

Observe that if x is any element of the group determined by 9.5, then $x^2 = e$.

例:

$$I = (Z_4, \oplus), f_i: Z_4 \rightarrow Z_4, f_i(x) = (ix) \bmod 4, i = 0, 1, 2, 3$$

函数	导出的同余关系
$f_0(x)=0, x=0,1,2,3$	全域关系
$f_1(x)=x, x=0,1,2,3$	恒等关系 I_{Z_4}
$f_2(0)=f_2(2)=0,$ $f_2(1)=f_2(3)=2$	$\{<0,2>, <2,0>, <1,3>, <3,1>\} \cup I_{Z_4}$
$f_3(0)=0, f_3(1)=3,$ $f_3(2)=2, f_3(3)=1$	恒等关系 I_{Z_4}

该例子中, 每个同态可以导出一个同余关系, 若同态像一样, 则导出相同的同余关系

5-50 设 f_1, f_2 都是从代数系统 $\langle A, \star \rangle$ 到代数系统 $\langle B, * \rangle$ 的同态。设 g 是从 A 到 B 的一个映射使得对任意 $a \in A$, 都有

$$g(a) = f_1(a) * f_2(a)$$

证明: 如果 $\langle B, * \rangle$ 是一个可交换半群, 那么 g 是一个由 $\langle A, \star \rangle$ 到 $\langle B, * \rangle$ 的同态。

【5-8.(4)】

证明 因为对于任意的 $a, b \in A$, 都有

$$\begin{aligned} g(a \star b) &= f_1(a \star b) * f_2(a \star b) = f_1(a) * f_1(b) * f_2(a) * f_2(b) \\ &= f_1(a) * f_2(a) * f_1(b) * f_2(b) = g(a) * g(b) \end{aligned}$$

所以, g 是 $\langle A, \star \rangle$ 到 $\langle B, * \rangle$ 的同态。

Products and Quotients of Groups

积群和商群

Theorem 1

- If
 - G_1 and G_2 are groups,
- then
 - $G = G_1 \times G_2$
- is a group with binary operation defined by
 - $(a_1, b_1)(a_2, b_2) = (a_1 a_2, b_1 b_2)$.

Example 1

- Let G_1 and G_2 be the group Z_2 . For simplicity of notation, we shall write the elements of Z_2 as 0 and 1, respectively, instead of $\bar{0}$ and $\bar{1}$. Then the multiplication table of $G = G_1 \times G_2$ is given in Table.

$$G_1 = (Z_2, \oplus)$$

$$= \{0, 1\}$$

$$G_2 = (Z_2, \oplus)$$

$$= \{0, 1\}$$

$$G = G_1 \times G_2$$

Table 9.10 Multiplication Table of $Z_2 \times Z_2$

	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{1})$
$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{1})$
$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{0})$
$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{0})$

Example 1

- $|G|=4$, 所以 G 同构于 V (Klein) or Z_4
- Since G is a group of order 4, it must be isomorphic to V or to Z_4 , the only groups of order 4.
- The function
 - $f: V \rightarrow Z_2 \times Z_2$ defined by
 - $f(e) = (0, 0)$, $f(a) = (1, 0)$,
 - $f(b) = (0, 1)$, and $f(c) = (1, 1)$
- is an isomorphism.

Table 9.5

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Table 9.7

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

Table 9.6

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

Table 9.8

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Note-推广

- If we repeat Example 1 with Z_2 and Z_3 , we find that $Z_2 \times Z_3 \cong Z_6$.
- It can be shown, in general, that
 - $Z_m \times Z_n \cong Z_{mn}$
 - if and only if
 - $\text{GCD}(m, n) = 1$. — m, n 互质时 $Z_m \times Z_n$ 同构 Z_{mn}

定理1推广: $G_1, G_2, G_3 \dots G_n$ 是群, 则
 $G_1 \times G_2 \dots G_{n-1} \times G_n$ 也是群

例:

$$\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$$

$\mathbb{Z}_2 \times \mathbb{Z}_3$

$\oplus$	(0,0)	(0,1)	(0,2)	(1,0)	(1,1)	(1,2)
(0,0)	(0,0)	(0,1)	(0,2)	(1,0)	(1,1)	(1,2)
(0,1)	(0,1)	(0,2)	(0,0)	(1,1)	(1,2)	(1,0)
(0,2)	(0,2)	(0,0)	(0,1)	(1,2)	(1,0)	(1,1)
(1,0)	(1,0)	(1,1)	(1,2)	(0,0)	(0,1)	(0,2)
(1,1)	(1,1)	(1,2)	(1,0)	(0,1)	(0,2)	(0,0)
(1,2)	(1,2)	(1,0)	(1,1)	(0,2)	(0,0)	(0,1)

$\oplus$	[0]	[1]	[2]	[3]	[4]	[5]
[0]	[0]	[1]	[2]	[3]	[4]	[5]
[1]	[1]	[2]	[3]	[4]	[5]	[0]
[2]	[2]	[3]	[4]	[5]	[0]	[1]
[3]	[3]	[4]	[5]	[0]	[1]	[2]
[4]	[4]	[5]	[0]	[1]	[2]	[3]
[5]	[5]	[0]	[1]	[2]	[3]	[4]

Example 2

+	0	1
0	0	1
1	1	0

- Let $B = \{0, 1\}$ be the group, where $+$ is defined as above
- Then $B^n = B \times B \times \dots \times B$ (n factors) is a group with operation $\oplus$ defined by
 - $(x_1, x_2, \dots, x_n) \oplus (y_1, y_2, \dots, y_n)$
 - $= (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$.
- The identity of B^n is $(0, 0, \dots, 0)$, and every element is its own inverse.
-

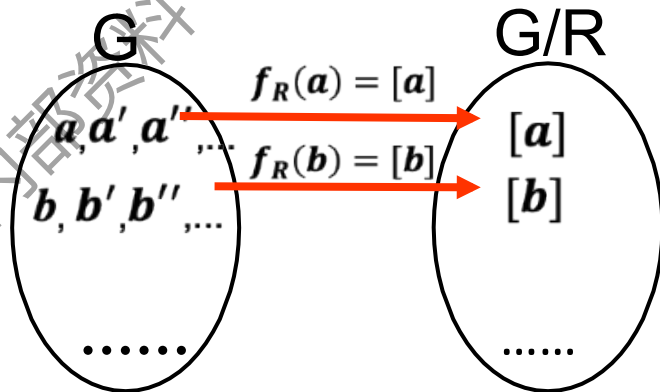
单位元(0, 0,..., 0), 每个元素是自己的逆

Theorem 2-商群

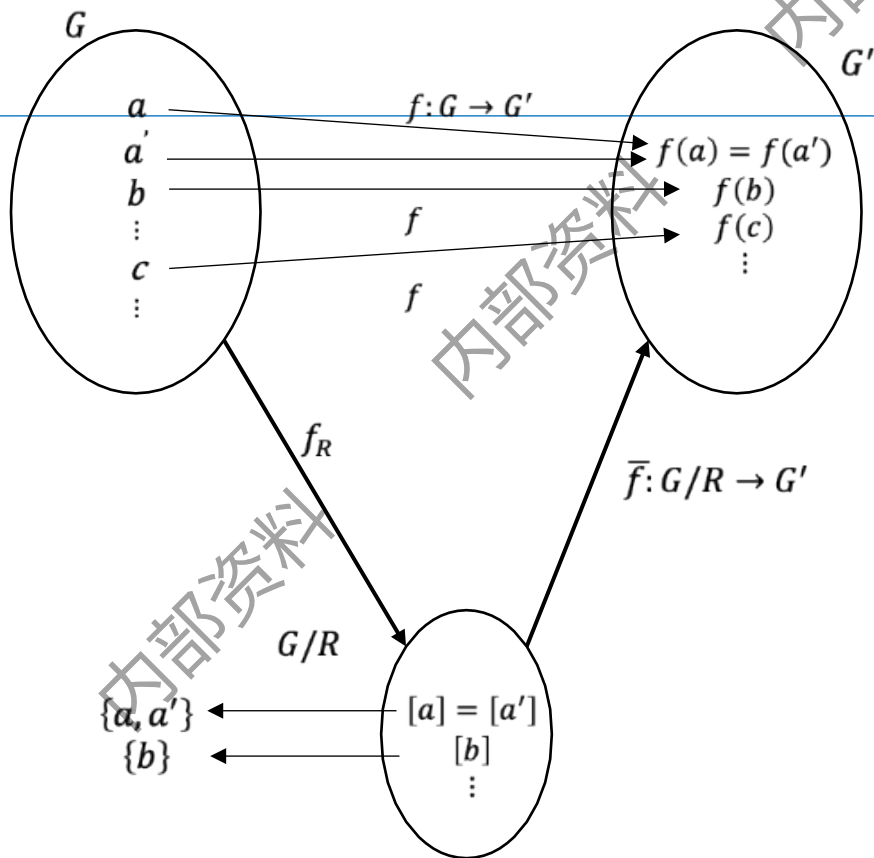
- Let R be a **congruence relation** on the group $(G, *)$.
- Then the semigroup $(G/R, \odot)$ is a group, where the operation $\odot$ is defined on G/R by
 - $[a] \odot [b] = [a * b]$
 - R 是群 G 上的同余关系, 则半群 $(G/R, \odot)$ 也是一个群。
- Proof: G 是群一定是独异点, 由9.3节**推论1**知 G/R 为独异点, 只需再证明每个元素有逆。
 - Since a group is a monoid, G/R is a monoid.
 - Let $[a] \in G/R$.
 - Then $[a^{-1}] \in G/R$, and
 - $[a] \odot [a^{-1}] = [a * a^{-1}] = [e]$.
 - So $[a]^{-1} = [a^{-1}]$.
 - Hence $(G/R, \odot)$ is a group.

Corollary 1

- If R is a congruence relation on a group G , then the function
 - $f_R: G \rightarrow G/R$
- given by
 - $f_R(a) = [a]$
- is a group homomorphism.



R 是群 G 上的一个同余关系，那么由 $f_R(a) = [a]$ 给出的函数 $f_R: G \rightarrow G/R$ 是一个同态



陪集

- 陪集 (coset) 定义及性质

Definition—陪集定义

内部资料

Let

H be a subgroup of a group G

$a \in G$

The **left coset**(左陪集) of H in G determined by a is the set

$$aH = \{ah \mid h \in H\}$$

The **right coset**(右陪集) of H in G determined by a is the set

$$Ha = \{ha \mid h \in H\}$$

A subgroup H of G is **normal**(正规子群) if

$$aH = Ha, \text{ for all } a \text{ in } G$$

H 是群 G 的**正规子群**, $\forall a \in G$, aH 是 G 中 H 的左陪集,

Ha 是右陪集。 a 是陪集的代表元

Definition—陪集定义

例 $(\mathbb{Z}_6, \oplus)$, 子群 $H = \{[0], [2], [4]\}$

$$[0] H = \{[0] \oplus [0], [0] \oplus [2], [0] \oplus [4]\} = \{[0], [2], [4]\}$$

$$[1] H = \{[1] \oplus [0], [1] \oplus [2], [1] \oplus [4]\} = \{[1], [3], [5]\}$$

$$[2] H = \{[2] \oplus [0], [2] \oplus [2], [2] \oplus [4]\} = \{[0], [2], [4]\}$$

$$[3] H = \{[3] \oplus [0], [3] \oplus [2], [3] \oplus [4]\} = \{[1], [3], [5]\}$$

$$[4] H = \{[4] \oplus [0], [4] \oplus [2], [4] \oplus [4]\} = \{[0], [2], [4]\}$$

$$[5] H = \{[5] \oplus [0], [5] \oplus [2], [5] \oplus [4]\} = \{[1], [3], [5]\}$$

$$[0] H = [2] H = [4] H$$

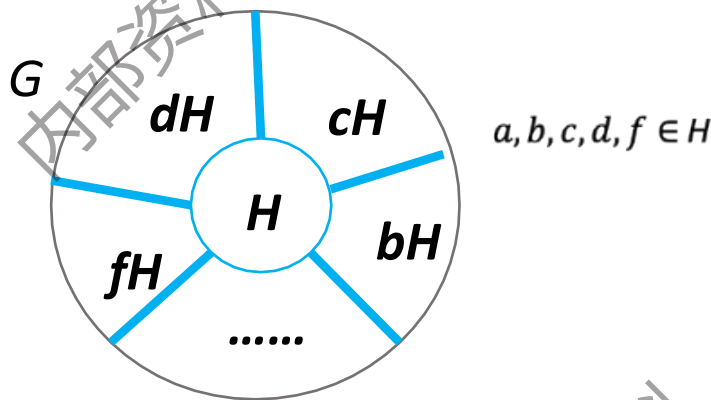
$$[1] H = [3] H = [5] H$$

任意两个左陪集，要么相等，要么不相交
什么情况相等，什么时候相交？

陪集性质

子群 $H \subseteq G$, $\forall a, b \in G$

- (1) $aH=H$, 当且仅当 $\forall a \in H$
- (2) $aH=bH$, 当且仅当 $b \in aH$
- (3) $aH \cap bH = \emptyset$, 当且仅当 $b \notin aH$
- (4) $aH \cup bH \cup cH \dots = G$
- (5) $\forall a, b \in G$, 如果 $a^{-1}b \in H$, 则 aRb , 为等价关系



Example 3

- Let
 - G be the symmetric group S_3 .
 - The subset $H = \{f_1, g_2\}$ is a subgroup of G .
- Compute all the distinct left cosets of H in G .

计算 G 中的所有 H 的左陪集。其中 $H = \{f_1, g_2\}$, $G = S_3$

Example 3-1

$$H = \{f_1, g_2\}$$

•Solution

• If $a \in H$, then $aH = H$. Thus

• $f_1 H = g_2 H = H$.

• $f_2 H = \{f_2, g_1\}$

• $f_3 H = \{f_3, g_3\}$

• $g_1 H = \{g_1, f_2\} = f_2 H$

• $g_3 H = \{g_3, f_3\} = f_3 H$

*	f_1	f_2	f_3	g_1	g_2	g_3
f_1	f_1	f_2	f_3	g_1	g_2	g_3
f_2	f_2	f_3	f_1	g_3	g_1	g_2
f_3	f_3	f_1	f_2	g_2	g_3	g_1
g_1	g_1	g_2	g_3	f_1	f_2	f_3
g_2	g_2	g_3	g_1	f_3	f_1	f_2
g_3	g_3	g_1	g_2	f_2	f_3	f_1

•The distinct left cosets of H in G are

• $H, f_2 H$, and $f_3 H$.

Example 3-2

$H = \{f_1, g_2\}$ ，求右陪集？

• Solution

• If $a \in H$, then $Ha = H$. Thus

• $Hf_1 = Hg_2 = H = \{f_1, g_2\}$.

• $Hf_2 = \{f_2, g_3\}$

• $Hf_3 = \{f_3, g_1\}$

• $Hg_1 = \{g_1, f_3\} = Hf_3$

• $Hg_3 = \{g_3, f_2\} = Hf_2$

• 有些左右陪集不相等！

• $f_2H = \{f_2, g_1\}$

• $Hf_2 = \{f_2, g_3\}$

*	f_1	f_2	f_3	g_1	g_2	g_3
f_1	f_1	f_2	f_3	g_1	g_2	g_3
f_2	f_2	f_3	f_1	g_3	g_1	g_2
f_3	f_3	f_1	f_2	g_2	g_3	g_1
g_1	g_1	g_2	g_3	f_1	f_2	f_3
g_2	g_2	g_3	g_1	f_3	f_1	f_2
g_3	g_3	g_1	g_2	f_2	f_3	f_1

Example 3-3

$H = \{f_1, f_2, f_3\}$, 求左、右陪集?

• Solution

• $g_1 H = H g_1 = \{g_1, g_2, g_3\}$.

• $f_1 H = H f_1 = \{f_1, f_2, f_3\}$

• 左陪集=右陪集

• $aH = H a$

•

*	f_1	f_2	f_3	g_1	g_2	g_3
f_1	f_1	f_2	f_3	g_1	g_2	g_3
f_2	f_2	f_3	f_1	g_3	g_1	g_2
f_3	f_3	f_1	f_2	g_2	g_3	g_1
g_1	g_1	g_2	g_3	f_1	f_2	f_3
g_2	g_2	g_3	g_1	f_3	f_1	f_2
g_3	g_3	g_1	g_2	f_2	f_3	f_1

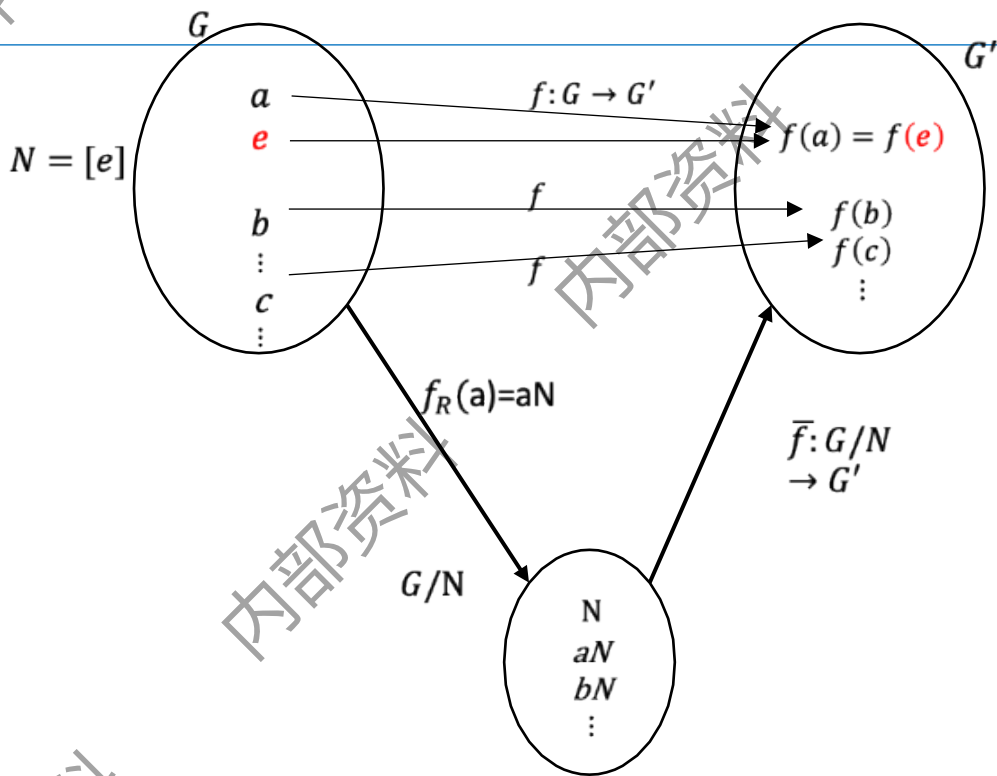
Theorem 3

- Let

- R be a congruence relation on a group G
- $H = [e]$, the equivalence class containing the identity.

- Then

- H is a normal subgroup of G
- $[a] = aH = Ha$, for each $a \in G$



Theorem 4

- Let

- N be a normal subgroup of a group G
- R be the following relation on G , aRb *if and only if* $a^{-1}b \in N$. (N 是正规子群, 如果 $a^{-1}b \in N$, 则 aRb)

- Then

- (a) R is a congruence relation on G .
- (b) N is the equivalence class $[e]$ relative to the identity of G .

前提

1. R 同余
2. $N=[e]$

结论

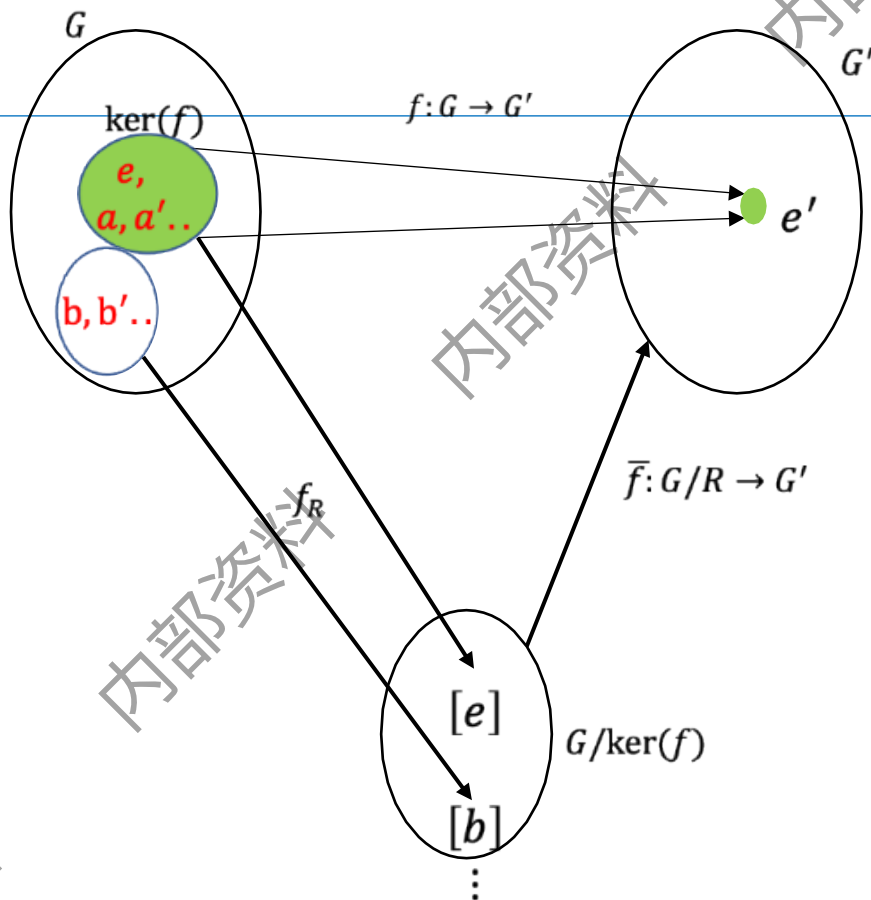
Corollary 2

- Let

- f be a homomorphism from a group $(G, *)$ **onto** a group $(G', *')$
- the **kernel**(核) of f , $\ker(f)$, be defined by
- $\ker(f) = \{a \in G \mid f(a) = e'\}$.

- Then

- $\ker(f)$ is a normal subgroup of G .
- The quotient group $G/\ker(f)$ is isomorphic to G' .



正规子群小结

- 1. 满足 $aH=Ha$, H 为正规子群
- 2. Abel 群的任何子群是正规子群
- 3. 群 G 中的同余关系 R , 确定的等价类 $N=[e]$
- 4. 同态核, $\text{Ker}(f)$

正规子群的判别 (补)

正规子群 $H \subseteq G$, 判别定理:

1. $aH = Ha, \forall a \in G$

2. $\forall a \in G, aHa^{-1} = H$

3. $\forall a \in G, h \in H, aha^{-1} \in H$

三个判别条件等价

Key ideas for review

Binary operation

Commutative, Associative

Semigroup, Monoid, Group

Subsemigroup, Submonoid, Subgroup

Isomorphism, Homomorphism

Congruence relation R on semigroup $(S,*)$

Quotient semigroup S/R ,

Order of group, S_n , Z_n , 置换群, 循环群

Left and right coset, Normal subgroup

5-10 设 $\langle S, * \rangle$ 是一个半群, $a \in S$, 在 S 上定义一个二元运算 $\square$, 使得对于 S 中的任意元素 x 和 y , 都有

$$x \square y = x * a * y$$

证明二元运算 $\square$ 是可结合的。

【5-3.(2)】

证明 因为

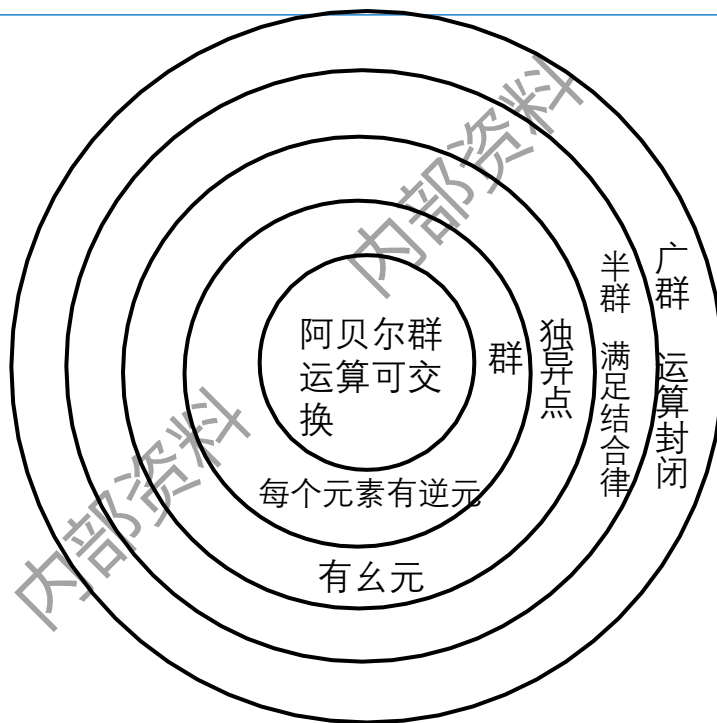
$$(x \square y) \square z = (x * a * y) \square z = (x * a * y) * a * z = x * a * y * a * z$$

$$x \square (y \square z) = x \square (y * a * z) = x * a * (y * a * z) = x * a * y * a * z$$

所以

$$(x \square y) \square z = x \square (y \square z)$$

$(S, *)$





Coding of Binary Information and Error Detection

- Encoding function (编码函数)
- Hamming distance (汉明距离)
- Group Codes (群码)
- Parity check matrix (奇偶校验矩阵)



Group Codes – 群码

- An (m, n) encoding function $e: B^m \rightarrow B^n$ is called a *group code* if
 - $e(B^m) = \{e(b) \mid e(b) \in B^n\} = \text{Ran}(e)$
- is a subgroup of B^n

Example 7: is e a group code?

- Consider the $(3, 6)$ encoding function $e: B^3 \rightarrow B^6$ defined by

$$e(000) = 000000$$

$$e(001) = 001100$$

$$e(010) = 010011$$

$$e(011) = 011111$$

$$e(100) = 100101$$

$$e(101) = 101001$$

$$e(110) = 110110$$

$$e(111) = 111010$$

code word

验证 B^3 是 B^6
的子群

Example 7: is e a group code?

- We must show that the set of all code words
 - $N = \{000000, 001100, 010011, 011111, 100101, 101001, 110110, 111010\}$
- is a subgroup of B^6 .

Constructing

maximum likelihood decoding function
associated with a given group code



Group code word 群码字

- Let $e: B^m \rightarrow B^n$ be an (m, n) encoding function that is a group code.
- Thus the set N of code words in B^n is a subgroup of B^n whose order is 2^m , say
 - $N = \{x^{(1)}, x^{(2)}, \dots, x^{(2^m)}\}$.

$$N \subset B^n$$



Coset leader – 陪集头

$$N = \{\bar{0}, x^{(2)}, x^{(3)}, \dots, x^{(i)}, \dots, x^{(j)}, \dots, x^{(2^m)}\}$$

- Suppose that the code word $x = e(b)$ is transmitted and that the word x_t is received.
- The left coset of x_t is
 - $x_t \oplus N = \{\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{2^m}\}$ where $\varepsilon_i = x_t \oplus x^{(i)}$
- if ε_i is a coset member with smallest weight, then $x^{(i)}$ must be a code word that is closest to x_t .
- An element ε_i having smallest weight, is called a *coset leader*.

Decoding table

译码表

- Constructing a decoding table, each row is a left coset of N with the first element $\epsilon^{(i)}$ the coset leader.

$\bar{0}$	$x^{(2)}$	$x^{(3)}$	...	$x^{(2^m-1)}$
$\epsilon^{(2)}$	$\epsilon^{(2)} \oplus x^{(2)}$	$\epsilon^{(2)} \oplus x^{(3)}$	...	$\epsilon^{(2)} \oplus x^{(2^m-1)}$
.	.	.		.
.	.	.		.
.	.	.		.
$\epsilon^{(2')}$	$\epsilon^{(2')} \oplus x^{(2)}$	$\epsilon^{(2')} \oplus x^{(3)}$	...	$\epsilon^{(2')} \oplus x^{(2^m-1)}$

Example 4

- Consider the (3, 6) group code
 - $N = \{000000,$
 - 001100,
 - 010011,
 - 011111,
 - 100101,
 - 101001,
 - 110110,
 - 111010}
 - $= \{x^{(1)}, x^{(2)}, \dots, x^{(8)}\}.$



Example 4

Constructing decoding table:

000000	001100	010011	011111	100101	101001	110110	111010
000001	001101	010010	111110	100100	101000	110111	111011
000010	001110	010001	011101	100111	101011	110100	111000





Example 4

Constructing decoding table:

$X=100101$ $x_t=000101, 100001$

•	=====						
•	000000	001100	010011	011111	100101	101001	110110 111010
•	-----						
•	000001	001101	010010	111110	100100	101000	110111 111011
•	000010	001110	010001	011101	100111	101011	110100 111000
•	000100	001000	010111	011011	100001	101101	110010 111110
•	010000	011100	000011	001111	110101	111001	100110 101010
•	100000	101100	110011	111111	<u>000101</u>	001001	010110 011010
•	000110	001010	<u>010101</u>	011001	100011	101111	110000 111100
•	010100	011000	000111	001011	110001	111101	100010 101110
•	=====						

换陪集头重排序





Simplified decoding technique

简化译码技术

- Suppose that the (m, n) group code is $e_H: B^m \rightarrow B^n$, where $\mathbf{H}$ is a given **parity check matrix**.

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{m \times r} \\ \mathbf{I}_r \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ \hline 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$





Simplified decoding technique

简化译码技术

- Then the function $f_H: B^n \rightarrow B^r$ defined by
 - $f_H(x) = x * H, x \in B^n$
- is a homomorphism from the group B^n to the group B^r .

见11.1定理5





Example 4

(3, 6)

$$H = \begin{bmatrix} H_{m \times r} \\ I_r \end{bmatrix} =$$

$$\begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ \hline 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

$B^r = \{$
 000
 001
 010
 100
 011
 101
 110
 111
 $\}$

```

• =====
• 000000 001100 010011 011111 100101 101001 110110 111010
• -----
• 000001 001101 010010 111110 100100 101000 110111 111011
• 000010 001110 010001 011101 100111 101011 110100 111000
• 000100 001000 010111 011011 100001 101101 110010 111110
• 010000 011100 000011 001111 110101 111001 100110 101010
• 100000 101100 110011 111111 000101 001001 010110 011010
• 000110 001010 010101 011001 100011 101111 110000 111100
• 010100 011000 000111 001011 110001 111101 100010 101110

```



Syndrome – 校验子

- It follows from Corollary I of Section 9.5 that B^r and B^n/N are **isomorphic**, where
 - $N = \{x \in B^n \mid x * H = 0\} = \ker(f_H) = e_H(B^n)$
- under the isomorphism
 - $g: B^n/N \rightarrow B^r$ defined by
 - $g(xN) = f_H(x) = x * H$
- The element $x * H$ is called the **syndrome** (特征值或校验子) of **x**

9.5 节推论1中, B^r 和 B^n/N 同构

Example 5

- Consider the parity check matrix and the (3, 6) group code $e_H: B^3 \rightarrow B^6$.

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \left. \begin{array}{l} e(000) = 000000 \\ e(001) = 001011 \\ e(010) = 010101 \\ e(011) = 011110 \\ e(100) = 100110 \\ e(101) = 101101 \\ e(110) = 110011 \\ e(111) = 111000 \end{array} \right\} \text{code word}$$

Example 5

- $N = \{000000, 001011, 010101, 011110, 100110, 101101, 110011, 111000\}$

Syndrome of Coset Leader	Coset leader
000	000000
001	000001
010	000010
011	001000
100	000100
101	010000
110	100000
111	001100

Example 5

- If $x_t = \mathbf{001110}$, then $f_H(x_t) = x_t * \mathbf{H} = 101$, same as $\varepsilon = 010000$.
- $x = x_t \oplus \varepsilon = 001110 \oplus 010000 = 011110 = e(011)$, so decode 001110 as 011.

Syndrome of Coset Leader	Coset leader
000	000000
001	000001
010	000010
011	001000
100	000100
101	010000
110	100000
111	001100

$$x_t * \mathbf{H} = \text{syndrome}$$